

VINCENT ONG

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EDUCATION

National University of Singapore, Singapore
Bachelor of Computing in Computer Science

— Present

SKILLS

Programming Languages: Java, Python, Go, Typescript, Javascript, SQL, R
Frameworks: React, Gin, Next.js, LangGraph, PyTorch, FastAPI, Express.js
Technologies: PostgreSQL, Docker, Git, AWS, Redis, GCP, AWS Bedrock

EXPERIENCE

Software Engineering Intern
Sonzai Labs

Mar 2025 — Jun 2025
Singapore

- Created a high-concurrency **clan system** for the startup's flagship game "Telemafia" using Go and **RESTful APIs**, reducing database load with **Redis** caching and atomic **ScyllaDB CQL** queries.
- Engineered **responsive and performant** game features for global users by building core gameplay **UI/UX** using **React**, **Typescript**, **Tailwind CSS**, **i18n** for internationalization and rapid bug resolution on a **CI/CD** workflow with QA team.
- **Led frontend development** for the secure launch and minting of **500 NFTs** by integrating **smart contracts** with Web3 wallet authentication and configuring **Cloudflare WAF** to mitigate DDoS and XSS attacks.
- Developed a leaderboard system to ensure **real-time ranking accuracy** by creating **Python** pipelines on **Windmill Orchestrator** and utilizing **Apache Pulsar** (Pub/Sub) and **ScyllaDB** for **automated data reconciliation**.

SAF Infantry 3rd Sergeant
Singapore Armed Forces

Mar 2023 — Jan 2025
Singapore

- Demonstrated **reliability** and **trustworthiness** in high-stakes environments, overseeing firearms **accountability** and taking **ownership** of the safety and operational readiness of recruits while **leading** as a section **commander**.

PROJECTS

ThreadTalk | Go, Gin, Next.js, Typescript, React, AWS, Docker, PostgreSQL

Dec 2025

- Architected a scalable, Reddit-inspired forum application by integrating **Go (Gin) RESTful APIs** with a Next.js, Typescript and React frontend, backed by a **PostgreSQL** relational database.
- Optimized data retrieval for **deep discussion threads** using recursive hierarchical comment nesting and infinite scroll with **concurrent processing** for speed.
- Automated secure production deployments by containerizing the serverless backend with **Docker** for **AWS Lambda**, establishing a **CI/CD** pipeline via **GitHub Actions** and **OIDC** for keyless **Amazon ECR** integration.
- Hardened application security and data integrity using **AWS VPC** isolation, **IAM** least-privilege policies and **JWT** authentication while automating **RDS** schema migrations with **Go**.

Veridien | Javascript, Express.js, Next.js, Node.js, PostgreSQL, Vercel, Railway

Jun 2025 — Jul 2025

- Built a full-stack **CRUD** journalling platform using Next.js, **Express.js** and PostgreSQL, promoting creative expression.
- Enabled seamless writing by implementing email **authentication** and real-time **autosave** to preserve user drafts.

Visente | Entrepreneurship

Nov 2022 — Mar 2023

- Generated approximately **\$100 profit** by launching a niche fragrance decanting venture, **selling 40+ bottles** by leveraging consumer demand for **affordable access to diverse scents**.
- Identified high-margin opportunities by executing strategic market research on **competitor pricing** and **consumer trends**, using **fragrance profiling** to optimize **cost efficiency** and production selection.

NUS Fintech Summit Hackathon | XRP Ledger SDK, Python, FastAPI

Jan 2026 — Jan 2026

- Architected **blockchain rails** on **XRP Ledger** via **FastAPI**, by **automating** credential verification and transaction signing.
- **Eliminated 100% pre-funding requirements** by deploying **XRPL escrows** and **agent-driven** credit underwriting, replacing idle **collateral** locks with efficient conditional **settlement**.

CERTIFICATIONS

Google Cloud Professional Machine Learning Engineer Certification
Google • Expires: Jun 2027

Jun 2025